



OAA AI Innovation Grant – System Architecture Self-Aware Room

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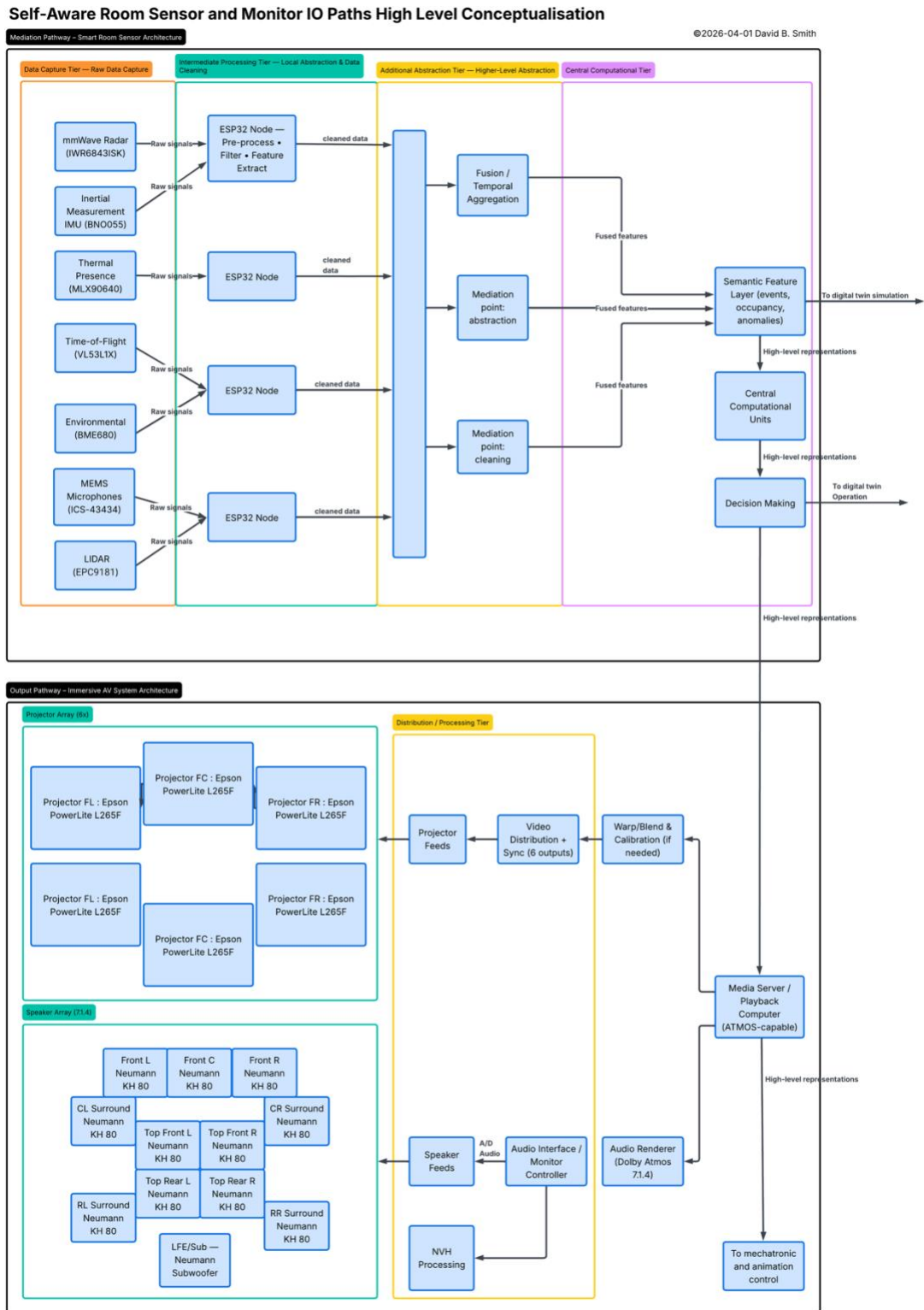
Introduction

This document describes the architectural framework for the Self-Aware Room system under the OAA AI Innovation initiative. The system is conceived as a multi-stage mediation architecture in which heterogeneous physical signals are progressively transformed into structured semantic representations capable of driving decision-making, simulation, and immersive media output.

The system is intentionally designed as an evolvable framework rather than a fixed installation. Its defining characteristics are the mediation pathways and transformation stages that govern how information moves through the system. Specific components—particularly at the input stage—are non-canonical and subject to modification as the system is deployed, evaluated, and refined.

1. Architectural Overview

1.1 System Diagram



At a high level, the system operates as a layered transformation pipeline in which signals pass through successive stages of abstraction. These stages move from raw physical measurement toward increasingly structured and interpretable representations. While these stages may be implemented across distributed hardware and software systems, they are best understood conceptually as four primary tiers: raw data acquisition, intermediate processing, higher-level abstraction, and semantic interpretation with decision output.

The architectural emphasis is not on any individual device or algorithm, but on the integrity and clarity of transitions between these tiers. Each transition represents a mediation boundary in which the form, meaning, and usability of the signal is transformed.

2. Data Capture Tier (Raw Data Acquisition)

The system begins with a heterogeneous set of sensing modalities operating in physical space. These sensors generate high-frequency, asynchronous, and device-specific data streams that reflect different aspects of the environment and its occupants. The current build is informed by a concrete bill of materials rather than a purely notional set of devices, providing a clearer picture of the system's sensing envelope while still allowing for evolution over time.

The present configuration includes the following sensing classes, instantiated by specific components in the current current procurement set:

- **Spatial and motion sensing:** mmWave radar evaluation boards (TI IWR6843), RealSense depth camera (D405), Time-of-Flight distance sensors (VL53L1X), and LiDAR-class depth capture via camera systems
- **Inertial sensing:** 9-DOF IMU modules (Bosch BNO055) providing acceleration, gyroscopic, and orientation data
- **Proximity and presence sensing:** VCNL4040 proximity sensors and short-range thermal presence sensors (Omron D6T series)
- **Thermal imaging:** MLX90640 thermal array sensors for coarse heat-mapped spatial awareness
- **Environmental sensing:** BME680 multi-sensor modules providing temperature, humidity, pressure, and gas/air quality metrics
- **Acoustic sensing:** digital MEMS microphones (ICS-43434) for ambient sound capture and analysis
- **Physiological sensing:** ECG front-end boards (ADS1292, AD8232), pulse sensors, and electrode interfaces for biosignal acquisition

These sensing systems are supported by distributed embedded platforms (ESP32-based development kits) that provide local connectivity, initial processing, and network integration. In addition to the specialized sensing systems listed above, more conventional

visual capture systems, including standard RGB cameras and webcams, are expected to play a significant role in the overall architecture. These systems provide high-resolution visual context that complements depth, thermal, and motion-based sensing, and will be integrated into the same mediation pipeline to support object recognition, scene interpretation, and behavioral analysis.

Although this list reflects the current implementation, it remains non-canonical. The architecture does not depend on any specific sensor, but on the system's capacity to ingest diverse data streams and transform them into a unified representational framework. As deployment proceeds, sensors may be added, removed, or reconfigured without altering the fundamental mediation structure of the system. As hardware and sensing capabilities advance, the system is expected to evolve through iterative upgrades, enabling higher resolution, broader modality coverage, and more sophisticated forms of environmental interpretation. In particular, emerging WiFi-based RF sensing systems present a promising direction, allowing for passive detection of occupancy, movement, and spatial dynamics through analysis of signal propagation and reflection patterns within the environment.

3. Intermediate Processing Tier (Local Abstraction and Cleaning)

Following acquisition, signals enter a stage of local processing typically handled by edge devices. At this stage, raw sensor outputs are conditioned into forms that are computationally tractable and comparable across modalities.

This process includes filtering, normalization, and preliminary feature extraction. The goal is not yet to interpret the data, but to remove noise, establish consistent scales, and reduce dimensionality where appropriate. The output of this stage consists of structured data streams, often represented as time-series vectors or compact feature sets.

This tier represents the first formal mediation boundary in the system. It converts device-dependent measurements into standardized computational representations, making subsequent cross-modal analysis possible.

4. Higher-Level Abstraction Tier (Fusion, Correlation, and Mediation)

The next stage integrates and reorganizes data across multiple sensor streams. Here, the system begins to construct a coherent representation of state by combining information that was previously isolated by modality and device.

This involves temporal alignment of asynchronous inputs, allowing signals captured at different rates and times to be meaningfully compared. It also includes sensor fusion, in which multiple modalities contribute to shared descriptors of system state.

A critical addition at this stage is the analysis of relationships among signals. Rather than treating each input independently, the system examines correlations across time, modality, and spatial distribution. These correlations may reveal patterns that are not visible within any single data stream, such as coordinated movement, environmental coupling, or emergent behavioral structures.

The mediation processes at this level—abstraction, cleaning, and correlation—collectively transform the system from a collection of signals into an integrated representation of activity and condition. The output is a set of fused feature representations that encode system state in a form suitable for interpretation.

5. Central Computational Tier (Semantic Representation and Decision)

At the highest level of abstraction, the system constructs semantic representations from the fused feature data. These representations translate quantitative signals into interpretable constructs that can support reasoning and action.

The semantic layer may include constructs such as discrete events, occupancy states, and anomaly indicators. These are not direct measurements, but derived interpretations that summarize system behavior in a structured form.

These representations are then processed by central computational systems responsible for pattern recognition, state modeling, and inference. The outcome of this processing feeds into a decision layer that determines how the system responds.

Importantly, this tier functions as the system's cognitive layer. It is the point at which signals acquire operational meaning within the system, enabling downstream systems to act on them.

6. Output Pathways and Digital Twin Integration

The outputs of the semantic and decision layers are routed to a range of external systems. These include immersive audiovisual environments, media rendering systems, and—critically—the Digital Twin or equivalent virtual environment.

The Digital Twin represents a central extension of the system's architecture. It provides a structured virtual environment in which semantic states can be visualized, simulated, and analyzed. This enables not only representation of current conditions but also exploration of potential future states and system behaviors.

Outputs to immersive systems such as projection and spatial audio further translate semantic states into perceptual experiences. These outputs form part of a feedback loop,

influencing human perception and behavior, which in turn re-enters the system through the sensor layer.

7. Temporal Structure of the System

The system operates across multiple temporal regimes, each corresponding to different stages of processing and interpretation.

Raw sensor signals exist in real time, while fused representations emerge over short temporal windows. Semantic interpretations occupy a slightly longer temporal frame, reflecting aggregated system state. Beyond these, the architecture allows for the incorporation of deferred computational processes that operate outside real-time constraints.

This layered temporal structure is not incidental; it is fundamental to the system's ability to integrate diverse forms of computation and interpretation.

8. Asynchronous Computational Extensions (Quantum Pathways)

The architecture is designed to support the integration of asynchronous computational pathways, including quantum processing systems. These systems do not operate within the same temporal constraints as the real-time environment and must therefore be treated as parallel interpretive processes.

In this model, inputs to such systems are drawn from the fused or semantic layers rather than from raw sensor data. These inputs represent structured snapshots of system state that can be encoded and processed externally.

The outputs of these processes—typically probabilistic or distributional in nature—are reintegrated into the system through a reconciliation process. This process compares classical representations of system state with alternative computational interpretations, enabling analysis of coherence, divergence, and potential transformation strategies.

Quantum computation is therefore not treated as a real-time control mechanism, but as a means of generating alternative representations of system state that may inform or refine higher-level decision-making.

9. System Philosophy

The Self-Aware Room is defined not by its components, but by its transformations. Its architecture prioritizes the clarity and flexibility of mediation pathways over the stability of any individual technology.

This approach ensures that the system can evolve over time, incorporating new sensing modalities, processing techniques, and computational paradigms without requiring fundamental redesign. It also positions the system as a platform for ongoing research into the relationships between physical signals, computational representations, and interpretive frameworks.

In this sense, the system functions not only as an engineered environment, but as a structured space for investigating how meaning and action emerge from complex, mediated interactions.